Imma Moncusí Laso 1667196

P3: SIESTA C60

1. Obtenció energies

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HOMO E= -0.5370867161013535  eV
LUMO E= 0.5207515730176034  eV
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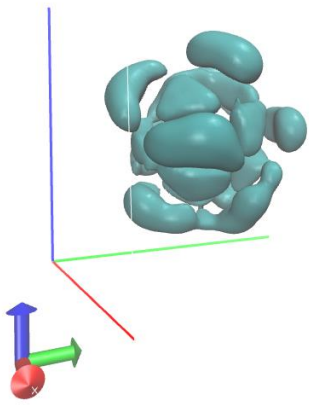
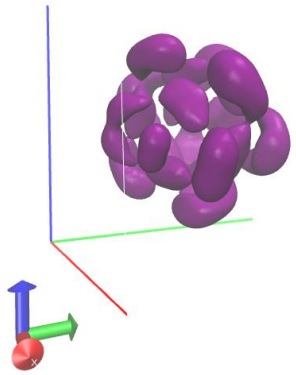
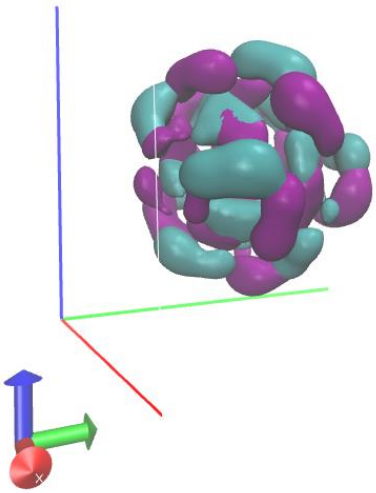
2. Representació VMD

HOMO (isovalue:0.017)	LUMO (isovalue:0.017)
	

HOMO i LUMO junts



3. Comparació amb bibliografia

Per obtenir la diferència absoluta entre els dos nivells els hem de restar.

ΔE=1.06 eV

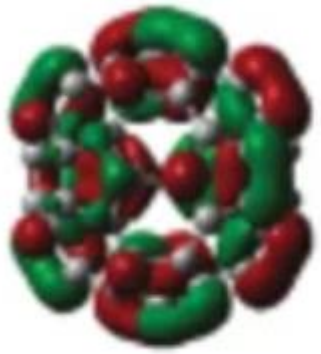
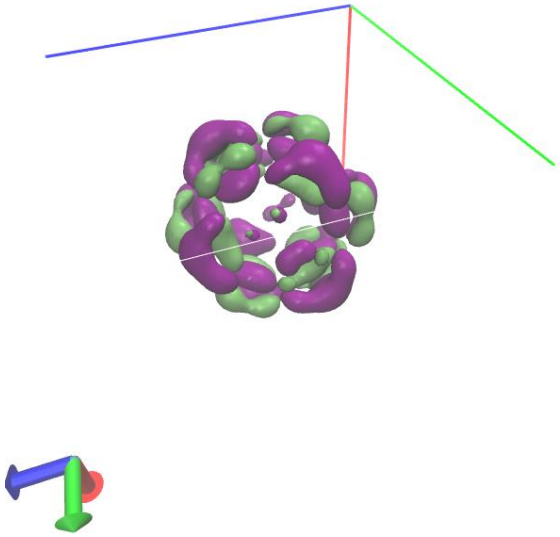
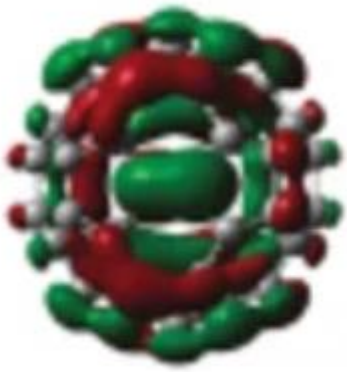
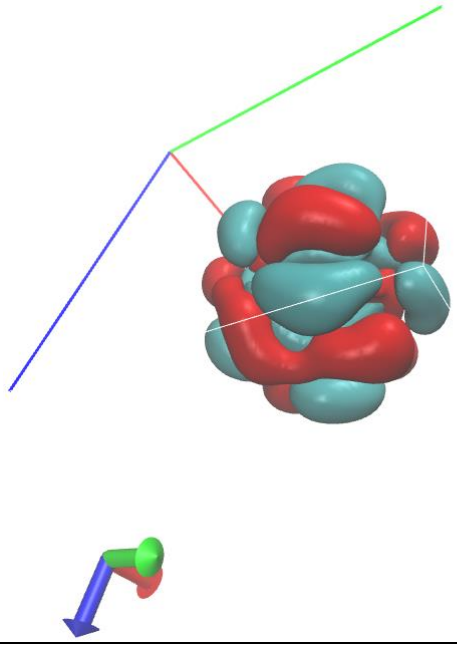
Referència 1: Wang, H., He, Y., Li, Y., & Su, H. (2012). Photophysical and electronic properties of five PCBM-like C60 derivatives: Spectral and quantum chemical view. *Journal of Physical Chemistry A, 116*(1), 255–262.

Table 3. LUMO Energy and HOMO−LUMO Energy Gap (E_{gap}) for F1, F2, F3, F4, F5, F6, F7, and C_{60} Obtained with the PBEPBE/6-311G(d,p) Calculations, along with the Predicted Open Circuit Voltages V_{oc}[a]

			LUMO/eV		V_{oc}/V	
samples	symm.	E_{gap}/eV	expt.	calc.	expt.	calc.
F1	C_1	1.456	−3.91[b]	−4.074	0.564[b]	0.526
F2	C_1	1.458	−3.91[b]	−3.992	0.571[b]	0.608
F3	C_1	1.452	−3.92[b]	−4.052	0.535[b]	0.548
F4	C_1	1.458	−3.90[b]	−4.013	0.596[b]	0.587
F5	C_1	1.455	−3.91[b]	−4.038	0.540[b]	0.562
F6	C_1	1.454		−4.018		0.582
F7	C_1	1.452		−4.033		0.567
C_{60}	I_h	1.650	−4.3[c]	−4.220		

[a] All data are compared to the experimental results if available. [b] Data cited from ref 35. [c] Data cited from refs 54 and 55.

LUMO R1	LUMO SIESTA

	
HOMO R1	HOMO SIESTA
	

No obstant, la referència està feta amb un altre mètode.

Si es busquen dades experimentals:

R2:Shibuta, M., Yamamoto, K., Ohta, T., Nakaya, M., Eguchi, T., & Nakajima, A. (2016). *Direct observation of photocarrier electron dynamics in C_{60} films on graphite by time-resolved two-photon photoemission*. **Scientific Reports, 6**, 35853.

ΔEcrystalline C60: 2.3-2.4 eV

ΔEoptic: 1.7-1.9 eV

4. Conclusió

Es pot observar que amb SIESTA i la referència (B3LYP) la diferència entre els gaps és apreciable, d'aproximadament 0.6 eV.

A primera vista els HOMO i LUMO s'assemblen. A simple ull és una mica complicat perquè la referència no dona les coordenades de visualització.